TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Procedure

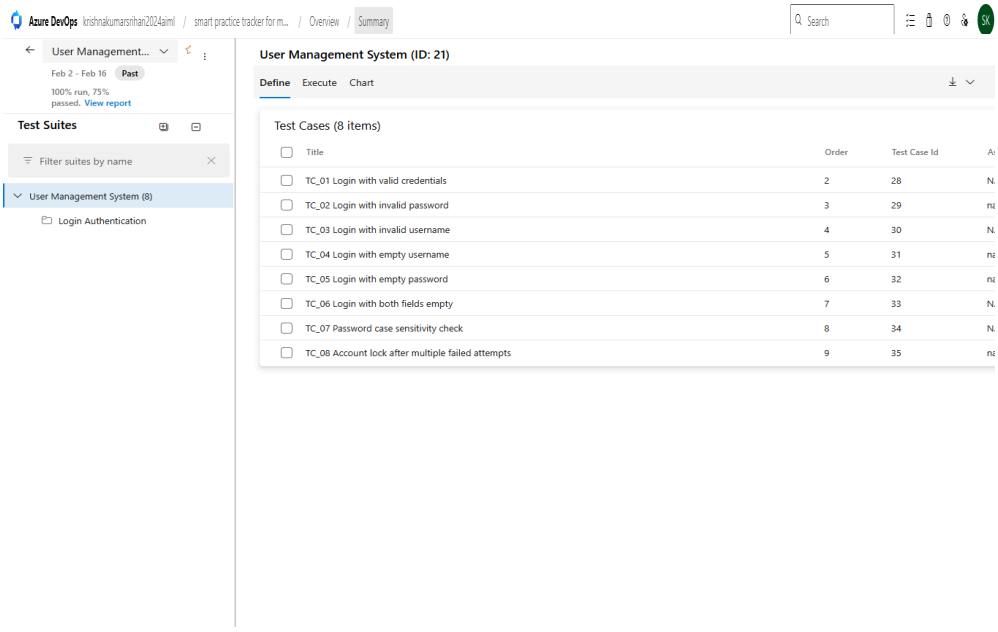
1. **Understand Core Features of the Application**
   - User Signup & Login
   - Viewing and Managing Playlists
   - Fetching Real-time Metadata
   - Editing playlists (rename, reorder, record)
   - Creating smart audio playlists based on categories (mood, genre, artist, etc.)
2. **Define User Interactions**
   - Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).
3. **Design Happy Path Test Cases**
   - Focused on validating that all features function as expected under normal conditions.
   - Example: User logs in successfully, adds item to playlist, or creates a category based playlist.
4. **Design Error Path Test Cases**
   - Simulate negative or unexpected scenarios to test robustness and error handling.
   - Example: Login fails with invalid credentials, save fails when offline, no recommendations found.
5. **Break Down Steps and Expected Results**
   - Each test case contains step-by-step actions and a corresponding expected outcome.
   - Ensures clarity for both testers and automation scripts.
6. **Use Clear Naming and IDs**
   - Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
   - Helps in quick identification and linking to user stories or features.
7. **Separate Test Suites**
   - Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

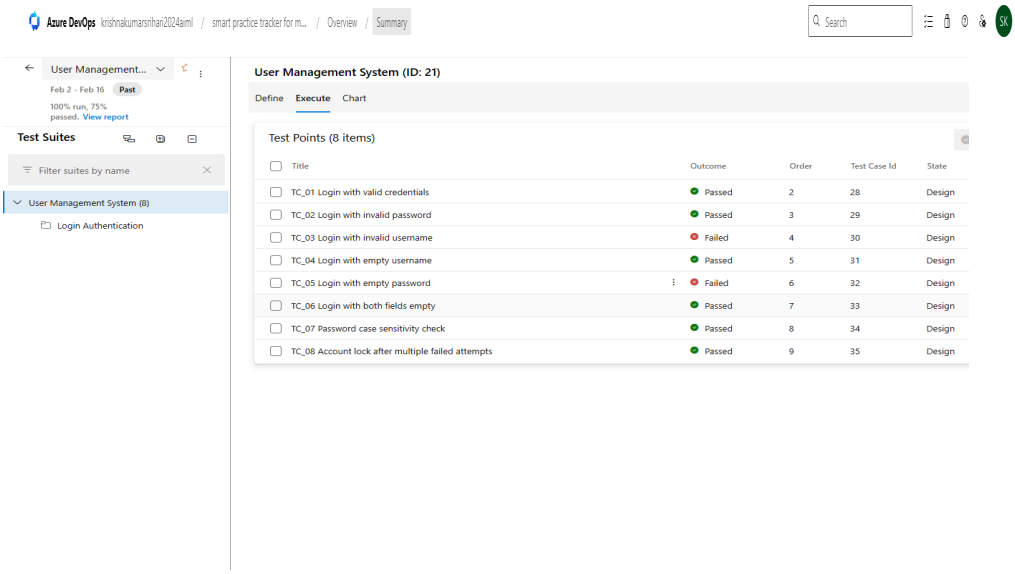
- Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

• **Test suite**





1.Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).
- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73). • As a user, I should be able to rename, record, and change the playlist (ID: 68). • As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. **TC01 – Successful Sign Up**
   - **Action:**
     - Go to the Sign-Up page.
     - Enter valid name, email, and password.
     - Click "Sign Up".
   - **Expected Results:**
     - Sign-Up form is displayed.
     - Fields accept values without error.
     - Account is created, and the user is redirected to the dashboard.
   - **Type**: Happy Path
2. **TC02 – Secure Login**
   - **Action:**
     - Go to the Login page.
     - Enter valid email and password.
     - Click on "Login".
   - **Expected Results:**
     - Login form is displayed.
     - Fields accept data without error.
     - User is logged in and redirected to the dashboard.
   - **Type:** Happy Path
3. **TC03 – Sign Up with Existing Email**
   - **Action:**
     - Go to the Sign-Up page.
     - Enter a name and an already registered email.
     - Click on "Sign Up".
   - **Expected Results:**
     - Fields accept data.
     - Error message "Email already registered" is displayed.
   - **Type:** Error Path
4. **TC04 – Login with Wrong Password**
   - **Action:**

        - Go to the Login page.
        - Enter valid email and incorrect password.
        - Click on "Login".
    - **Expected Results:**
        - Input is accepted.
        - Error message "Invalid username or password" is shown.
    - **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. **TC05 – View Playlist Page**
    - **Action:**
        - Log in successfully.
        - Navigate to "My Playlists" section.
    - **Expected Results:**
        - All created playlists are displayed clearly.
    - **Type:** Happy Path
2. TC06 – Playlist Loading Failure
    - **Action:**
        - Disconnect from the internet.
        - Navigate to "My Playlists".
    - **Expected Results:**
        - Network is offline.
        - Error message "Unable to load playlists" is shown.
    - **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. **TC07 – Real-Time Metadata Display**
    - **Action:**
        - Play a song.
        - Observe the metadata panel.
    - **Expected Results:**
        - Metadata (title, artist, album, duration) is displayed and updates in real time.
    - **Type:** Happy Path
2. TC08 – Metadata Not Updating
    - **Action:**
        - Play a different song.
        - Observe the metadata panel.
    - **Expected Results:**
        - Metadata remains static or shows default/fallback message.
    - **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. **TC09 – Rename Playlist Successfully**

- **Action:**
  - Navigate to "My Playlists".
  - Click "Rename" next to a playlist.
  - Enter a new name and click "Save".
- **Expected Results:**
  - Playlist name updates successfully.
- **Type:** Happy Path

2. TC10 – Rename with Blank Name

- **Action:**
  - Click "Rename" on a playlist.
  - Leave the field blank.
  - Click "Save".
- **Expected Results:**
  - Error message "Playlist name cannot be empty" is shown.
- **Type:** Error Path

3. TC11 – Change Playlist Order

- **Action:**
  - Opena playlist.
    - Drag and drop songs to reorder.
  - Click "Save".
- **Expected Results:**
  - Playlist order is updated and saved.
- **Type:** Happy Path

4. TC12 – Change Playlist Order Fails

- **Action:**
  - Login and go to “My Playlists”.
  - Select a playlist.
  - Go offline or simulate server error.
  - Reorder songs and click “Save Order”.
- **Expected Results:**
  - Error message: "Failed to update order. Please check your connection".
- **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

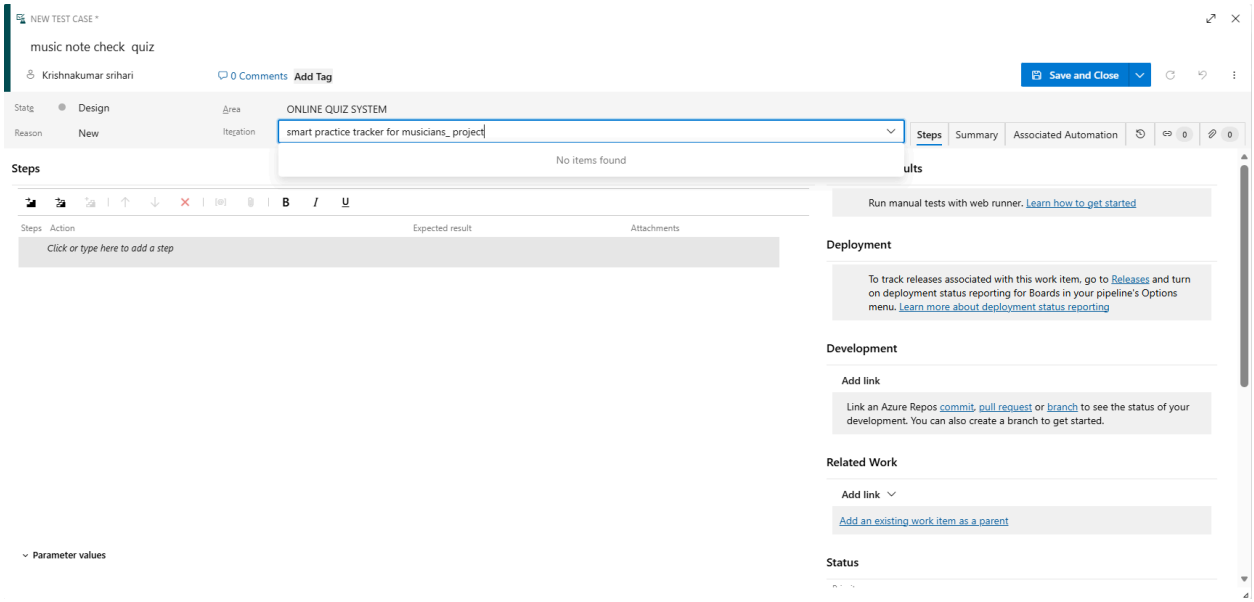
1. TC13 – Generate Playlist Based on Various Categories

- **Action:**
  - Login with valid credentials.
  - Click on "Generate Playlist".
  - Select categories.
  - Click "Generate Playlist".
- **Expected Results:**
  - Playlist is generated based on selected mood and categories.

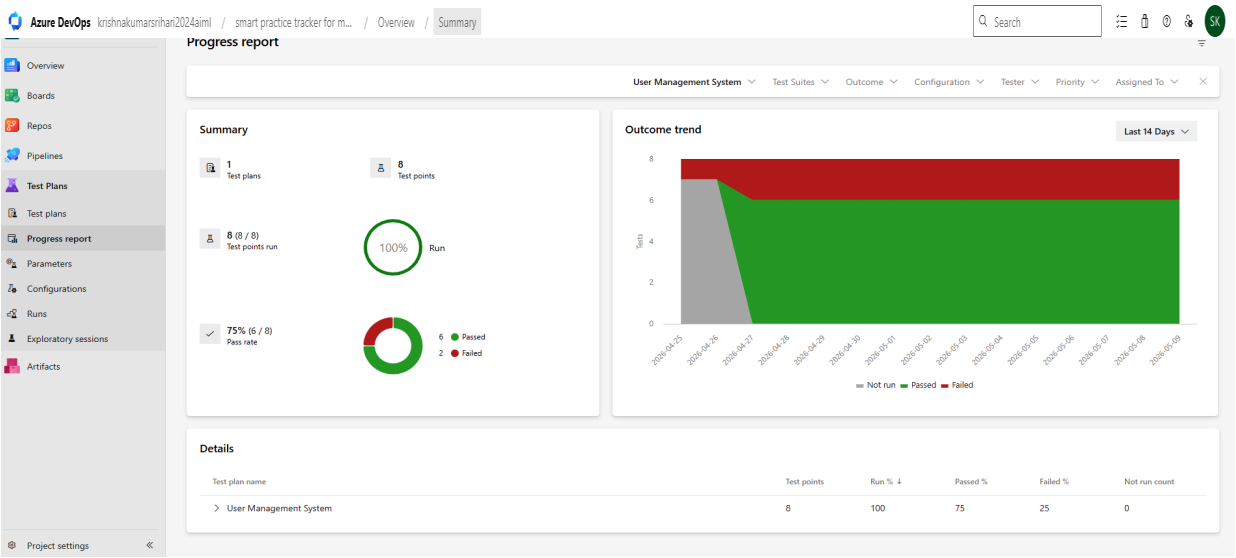
- **Type:** Happy Path

2. **TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input**
   - **Action:**
     - Login with valid credentials.
     - Click on "Generate Playlist".
     - Select categories.
     - Click "Generate Playlist".
   - **Expected Results:**
     - Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".
   - **Type:** Error Path

Test Cases



1.Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path